

SATYAM KUMAR

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SUMMARY

Software Engineer with 4+ years of experience in backend systems, distributed architecture, and cloud infrastructure. Currently at Microsoft building Azure PostgreSQL Flexible Server migration tooling — with deep expertise in PostgreSQL internals, replication, WAL, extensions, and Azure VM/networking. Strong C#, Go, and Java background with a track record of shipping high-impact features at scale. Targeting backend, cloud/database platform, and AI infrastructure roles. IIT Kanpur graduate.

TECHNICAL SKILLS

Core Languages: GoLang, Java, C#, Python, SQL

Backend & Systems: Spring Boot, .NET, Gin-Gonic, Node.js, REST APIs, Microservices, gRPC, Async processing

PostgreSQL & Databases: PostgreSQL (WAL, replication, CDC, pgoutput, pgcopydb, logical decoding, extensions), MySQL, SQL Server, Redis, DocumentDB

Messaging & Streaming: Apache Kafka, Zookeeper, AWS SQS/SNS — event-driven & pub-sub architectures

Cloud & Infrastructure: Azure (VMs, Networking, Flexible Server, Control/Data Plane), AWS (Lambda, EC2, SQS, SNS, CloudWatch), Docker, Kubernetes

Observability: Grafana, Prometheus — dashboards, alerting, incident response

WORK EXPERIENCE

Microsoft IDC

Hyderabad, India

Software Development Engineer — Azure PostgreSQL Migration

Feb 2025 – Present

- **Improved online migration success rate from 65% to 96%** by resolving critical incidents and edge-case bugs across single-to-flexible PostgreSQL server migrations, directly unblocking enterprise customers. [\[C#, PostgreSQL, Azure\]](#)
- **Boosted CDC phase performance by 3–4×** by integrating pgoutput, reducing replication overhead and cutting migration downtime significantly. [\[C#, pgoutput, CDC, PostgreSQL replication\]](#)
- **Made sync lag trustworthy** by revamping the lag calculator with a more robust algorithm, producing accurate values teams can confidently act on. [\[C#, PostgreSQL WAL\]](#)
- **Enabled Azure Linux support** by implementing multi-OS compatibility in the migration pipeline, replacing Ubuntu with Microsoft's faster in-house OS and reducing compute costs. [\[Azure VMs, Linux, C#\]](#)
- **Extended control-plane/data-plane communication** by engineering locale settings migration and extension allowlisting features — enabling the control plane to dynamically configure from the data plane, reducing manual customer work. [\[Azure, C#, distributed systems\]](#)
- **Patched two critical security vulnerabilities** — an MSRC privilege escalation granting unauthorized azure_su access, and an SSRF flaw enabling malicious internal API calls. [\[Security, Azure, C#\]](#)

A. P. Moller Maersk

Bangalore, India

Associate Software Engineer

Mar 2023 – Jan 2025

- **Architected core Inventory Management System** supporting real-time inventory tracking across 50+ global facilities with high data integrity. [\[Java, Spring Boot, Stored Procedures\]](#)
- **Reduced operational manpower by 40%** by automating Appointment, Dock, Gate, and Yard workflows end-to-end. [\[C#, .NET\]](#)
- **Designed a Multi-Role authentication system** supporting 10+ role-based access control levels, enforcing fine-grained API authorization for global teams. [\[Java, Spring Security\]](#)
- **Enabled global scalability** via Multi-Site functionality — onboarding 50+ worldwide facilities without architectural changes. [\[Distributed systems, Java\]](#)
- **Built observability stack** with real-time API dashboards and alerting, cutting MTTD for production incidents. [\[Grafana, Prometheus\]](#)

Siply Services Pvt Ltd

Bangalore, India

Software Engineer

Sep 2021 – Feb 2023

- **Led rewards system serving 300,000+ users** delivering a high-throughput, low-latency platform with zero major incidents post-launch. [\[GoLang, Node.js\]](#)

- **Built AutoPay for 20,000+ users** streamlining EMI payments, reducing failures and manual intervention. [\[GoLang\]](#)
- **Increased user engagement by 13%** via real-time event tracking and notifications. [\[AWS SNS/SQS, GoLang\]](#)
- **Reduced fraud for 15,000+ users** by integrating CKYC and Penny Drop KYC verification. [\[GoLang, KYC APIs\]](#)
- **Increased sell-flow success by 30%** by re-architecting to asynchronous processing, eliminating timeout failures under peak load. [\[GoLang, async systems\]](#)

ZestMoney

Remote

Software Engineer Intern

May 2020 – Jul 2020

- **Built high-throughput Pub-Sub messaging system** forming the backbone of real-time event processing. [\[GoLang, Kafka, Zookeeper, Docker\]](#)
- **Developed full KYC user flow** improving onboarding compliance and reducing drop-off. [\[ReactJS\]](#)

PROJECTS

- **Scalable OAuth Provider**
 - **Architected and implemented** a high-performance OAuth provider with focus on high availability and fault tolerance. [\[GoLang, Fiber, Gorm, PostgreSQL, Redis\]](#)
- **Praguna EV — Shell Eco Marathon**
 - **Calculated chassis dimensions** using Matlab to ensure compliance with 20+ Shell Eco Marathon competition rules for an energy-efficient electric prototype vehicle.
 - **Led vehicle dynamics design**, reducing air drag and weight through material and structural analysis. [\[SolidWorks, Ansys\]](#)

RELEVANT COURSES

UG Courses: Modern Cryptography, Introduction to Programming, Linear Algebra, Probability and Statistics, Machine Learning for WC

Online Courses: High Level System Design, Data Structures & Algorithms, Docker & Kubernetes, Low Level Design

EDUCATION

Indian Institute of Technology, Kanpur*Kanpur*

B.Tech in Electrical Engineering Jul 2017 – Jun 2021

JEE Advanced 2017: AIR 1524 — Top 0.5% nationally among 1.2M+ candidates

INTERESTS & SPORTS

- **Interests:** Business case studies, Cooking
- **Sports:** Chess, Badminton, Tennis, Table Tennis